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Plug-and-play fine grained neural cognitive diagnosis framework

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ABSTRACT

Cognitive diagnosis (CD) is a core task in intelligent education, which accurately assesses students' mastery of specific knowledge concepts (KCs) by analyzing their answer records. However, existing methods mainly rely on explicit interaction data and use diagnostic models for automatic knowledge proficiency inference. These methods lack systematic optimization for fine-grained knowledge level representation, making it difficult to fully reflect students' true learning status. To address this, this paper introduces a *Plug-and-Play **F**ine Grained **N**eural **C**ognitive **D**iagnosis Framework **(FNCD)** with Knowledge-Level Constraint Awareness*. The framework combines a knowledge proficiency evaluation module with students' answer records and a Q-matrix to statically assess knowledge mastery. It uses a student similarity construction method based on random grouping to reveal latent learning pattern associations. Additionally, it employs a multi-scale relational learning strategy and a Top-k attention-enhanced graph network mechanism to dynamically adjust the student similarity relationship network, accurately modeling the complex learning relationships between students. Ultimately, a joint training mechanism is used to optimize the outputs of each module, significantly improving the rationality, interpretability, and accuracy of CD. The experimental results demonstrate that FNCD, as an artificial intelligence-driven plug-and-play module, can be effectively integrated into existing CD models to enhance the modeling of fine-grained knowledge mastery and improve diagnostic accuracy, showcasing the application potential of artificial intelligence in personalized education.

1. Introduction

Cognitive Diagnosis (CD) plays a crucial role in educational data mining by analyzing students' responses to reveal mastery of specific knowledge concepts (KCs) (Wang et al., 2024b,a; Cheng et al., 2019). As intelligent education technologies become more popular, demand for CD to improve learning outcomes and support personalization grows (Yao et al., 2023). CD analyzes student-task interaction data to model knowledge levels and assess learning state. This helps students track progress and improve efficiency (as shown in Fig. 1(a)). CD is also widely applied in various fields, such as online gaming (Chen and Joachims, 2016), and intelligent medical diagnosis (Guo et al., 2017).

Cognitive Diagnosis Models (CDMs) have continuously developed. Their effectiveness depends on how well interaction functions model the complex relationships between students and KCs. While traditional Item Response Theory (IRT) (Lord, 1952) and Multidimensional Item Response Theory (MIRT) (Ackerman, 2014) provide the theoretical foundation for constructing CDMs, the assumptions underlying these methods limit their applicability. The DINA model (De La Torre, 2009), as a core model of CDM, allows for in-depth analysis of students' mastery of different KCs, thereby supporting the development of personalized learning strategies. In recent years, innovative research integrating deep neural networks (Wang et al., 2022; Yao et al., 2023;

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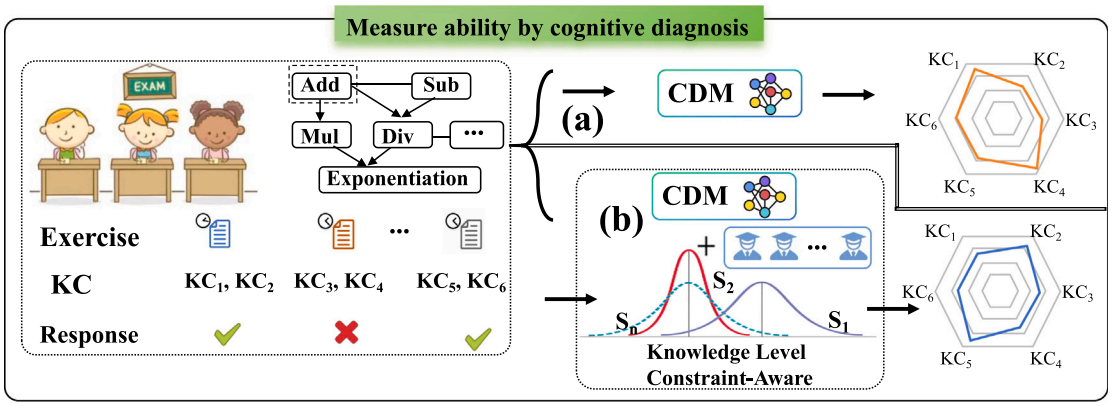


Fig. 1. A toy example of cognitive diagnosis, where CDM stands for the existing CD model. The upper part (a) represents the traditional CD approach, while the lower part (b) illustrates the CD approach proposed in this paper.

Wang et al., 2024a) has significantly enhanced the performance of CD models, driving the advancement of this field.

Although existing CD frameworks enhance diagnostic accuracy and interpretability to some extent through students' explicit interaction records (such as KCs (Ghosh et al., 2020; Sun et al., 2024), exercise texts (Sun et al., 2025e)), and exercise relationships, these methods primarily focus on automatic inference of knowledge proficiency based on diagnostic models. They lack systematic exploration of optimizing knowledge proficiency representation. We believe that this limitation may lead to significant discrepancies between the inferred knowledge proficiency and students' true abilities during fine-grained knowledge capability reasoning. Therefore, how to effectively integrate patterned information from students' past learning behaviors to construct a more precise model for knowledge proficiency representation, enabling accurate inference of students' true mastery level, remains a key challenge that needs to be addressed (as shown in Fig. 1(b)). Solving this challenge will not only advance the research in the field of CD but also substantially enhance the fairness, transparency, and interpretability of modern educational assessments.

To more accurately diagnose students' fine-grained cognitive state, this paper addresses key challenges in incorporating fine-grained knowledge constraint features. **First**, how to effectively quantify knowledge proficiency based on students' past learning behaviors is fundamental to addressing CD issues. **Second**, how to dynamically adjust similarity calculations based on prior statistical information, especially in the context of constantly changing learning patterns among student groups, remains a difficult problem due to the required flexibility. **Third**, the relationships between students are typically complex and high-dimensional. Capturing these relationships effectively within a fine-grained knowledge-level framework, while avoiding the interference of informational noise, is the core challenge in building a stable learning effect transfer mechanism. **Finally**, individual differences in students' abilities are significant. How to precisely capture these differences during the diagnostic process and construct inherent associative mechanisms is still a critical research direction for improving CD effectiveness.

To address the aforementioned challenges, we propose a Fine-Grained Cognitive Diagnosis Framework with Knowledge-Level Constraint Awareness, which significantly enhances the accuracy and interpretability of CD through several innovative strategies. **First**, the framework introduces a knowledge proficiency assessment module, which combines students' performance on tasks with a Q-matrix to statically evaluate their mastery of various KCs, laying the foundation for accurately capturing dynamic changes in knowledge proficiency. **Second**, a student similarity construction method based on random grouping is employed to analyze students' learning behaviors and compute similarities, thereby revealing potential learning pattern associations between groups. **Third**, the framework incorporates a multi-scale relational learning strategy, leveraging a Top-k attention-enhanced graph network mechanism. This allows for dynamic adjustment of student similarities while iteratively refining the learning relationship network, significantly improving the model's ability to represent learning patterns. **Finally**, a diagnostic enhancement strategy integrates the outputs of all modules, further optimizing the accuracy and interpretability of CD. These innovative strategies are organically combined to provide an effective solution for achieving more precise and efficient fine-grained CD.

In summary, we propose solutions to the above challenges and provide a more reasonable method for constraining the changes in students' knowledge state. The main contributions of this study are as follows:

- We propose an effective plug-and-play CD framework that constrains students' knowledge state changes[3], making it more aligned with real educational contexts.
- We introduce a fine-grained knowledge-level dynamic modeling method to accurately capture changes in students' knowledge level and learning relationships.
- We adopt a joint training mechanism to improve the accuracy and interpretability of CD, supporting applications in complex learning scenarios.

2. Related works

Accurately identifying students' knowledge state in real educational contexts is a highly valuable task. Subsequently, we will introduce relevant research from two perspectives: cognitive diagnosis and knowledge constraints.

2.1. Cognitive diagnosis

Cognitive Diagnosis (CD) has been widely applied in various fields, such as education, healthcare, and gaming, with the goal of analyzing learners' test data to uncover their potential traits and cognitive abilities. These insights help characterize the interaction between learners and tasks, providing theoretical support for personalized teaching and assessment (Zhou et al., 2021; Zhang et al., 2024).

Item Response Theory (IRT) is one of the early models in CD, aiming to establish a mathematical relationship between learners' abilities and

[3] To support research reproducibility, we have made the data and code publicly available at: https://github.com/xinjiesun-ustc/FNCD, encouraging further innovation in the field.

their task performance. IRT provides a statistical framework for analyzing individual differences by parameterizing task characteristics (such as difficulty and discrimination) and learners' latent abilities (Lord, 2012; An and Yung, 2014). However, IRT is limited in modeling cognitive traits along a single dimension, which proves insufficient for more complex scenarios. **We propose an FNCD framework integrated with IRT, creating a novel FNCD-IRT CD model.**

To overcome the limitations of unidimensional modeling, Multidimensional Item Response Theory (MIRT) was developed to represent learners' cognitive traits as multidimensional vectors, offering a more accurate reflection of their knowledge mastery (Ackerman, 2014). MIRT is widely used in educational assessments to measure learners' proficiency across multiple KCs, as it can better capture complex cognitive characteristics (Chalmers, 2012; Finch, 2010; Jewsbury and van Rijn, 2020). **Our FNCD framework further integrates with MIRT, leading to the development of a new FNCD-MIRT CD model.**

The DINA model assumes that completing a task requires mastery of all relevant KCs, with performance affected by noise, resulting in binary outcomes (mastered or not mastered) (Yamaguchi and Okada, 2020; De La Torre, 2009). This model has been crucial in personalized learning and knowledge assessment in educational practice (Tian et al., 2023). As research has advanced, extended DINA models have introduced multidimensional capabilities and integrated deep learning techniques for greater flexibility (Chen et al., 2021; Wang et al., 2024b). **Building on this, we integrate the FNCD framework with DINA, resulting in a new FNCD-DINA CD model.**

With the rapid development of machine learning technologies, CD has increasingly incorporated more intelligent approaches. Researchers have built learner ability prediction models using collaborative filtering and matrix factorization techniques, improving task performance prediction accuracy (Thai-Nghe et al., 2010; Thai-Nghe and Schmidt-Thieme, 2015; Toscher and Jahrer, 2010). Additionally, neural network-based models have been applied to train data attributes hierarchically, optimizing learning algorithms to improve classification performance and stability (Gierl et al., 2008; Dai et al., 2024).

The application of modern deep learning in CD has driven significant advancements in the field. For instance, NeuralCDM, through its nonlinear feature mapping, effectively captures the complex interactions between learners and tasks. The DIRT model cleverly combines the strengths of deep learning and traditional IRT, enhancing the model's predictive capability while maintaining high interpretability (Cheng et al., 2019; Wang et al., 2020). Moreover, many deep learning-based CDMs built on NeuralCDM further enhance the model's diagnostic performance and interpretability by introducing additional parameters or structures. For example, Cheng et al. (2021) add a KC importance vector to the input layer of NeuralCDM, significantly improving the model's overall performance. Ma et al. (2022) propose the KSCD model, which introduces additional parameters for KCs, better capturing the relationships between students, tasks, and KCs.

At the same time, Li et al. (2024) proposes the ID-CDF framework based on NeuralCDM, using a Response-Ability-Response paradigm to improve diagnostic accuracy and avoid overfitting. DCD (Chen et al., 2024) integrates decoupling and finite label alignment modules to tackle the challenges posed by limited annotated data. Recent studies have extended NeuralCDM into the ICD model (Qi et al., 2023), which employs a three-layer neural network to separately model the influence of exercises on knowledge concepts, the interactions among concepts, and the influence of concepts on exercises. This design specifically addresses the issues of concept interaction and quantitative exercise–concept relationships, thereby enhancing interpretability and predictive performance. In contrast, the HCD model (Sun et al., 2025a) focuses on hierarchical constraints in knowledge structures. It introduces a hierarchy mapping layer and a hierarchy-aware convolutional attention mechanism to capture both intra- and inter-level differences in student abilities. By emphasizing group-level hierarchical constraints, HCD improves the fairness and scientific rigor of cognitive diagnosis. These models differ significantly from our work, which introduces fine-grained, constraint-aware features at the knowledge level to achieve more accurate and individualized student diagnosis. **We combine the FNCD framework with NeuralCDM to propose a new FNCD-NCDM CD model.**

2.2. Knowledge constraint

In recent years, the integration of knowledge constraints into machine learning methods has become an increasingly important research direction (Ganchev et al., 2010; Hu et al., 2016; Ziad et al., 2024). The incorporation of such constraint-based knowledge not only facilitates the learning process but also significantly improves the model's generalization ability and interpretability. In deep neural networks, a common approach is to design specialized network architectures or features for specific tasks (Sun et al., 2025b). However, this approach is often inherently limited to specific tasks, models, or knowledge domains, thereby restricting its broad applicability.

Research has shown that cognitive development is largely guided by a set of complex constraints specific to a domain, which limit the types of naturally learnable structures within that domain Keil (1981). In the medical field, particularly in dental automation, the integration of domain knowledge has shown significant potential. For example, Zheng et al. (2020) propose a Dense U-Net with anatomical constraints, which incorporates oral anatomical knowledge to improve the model's performance and achieve significant results in situations with sparse data.

In the field of natural language processing, the rise of large-scale pre-trained models has made the integration of knowledge especially critical. Knowledge provides statistical models with common sense and logical capabilities, significantly boosting language understanding and generation (Zhu et al., 2023). At the same time, constraint optimization in neural networks has received widespread attention to improve model robustness and performance. Kervadec et al. (2022) propose using a logarithmic barrier expansion method to handle a large number of constraints, effectively enhancing constraint optimization performance. Additionally, Hu et al. (2018) combine maximum entropy Inverse Reinforcement Learning (IRL) with the PR framework, proposing an efficient algorithm to integrate large-scale constraints in generative models, optimizing constraint learning objectives with strong practicality and innovation. **Our FNCD achieves accurate assessment of students' knowledge state by introducing constraint optimization of knowledge level in neural networks.**

In the field of education, modeling students' knowledge state is critical for personalized education and support. Wang et al. (2023) enhance the interpretability and predictive accuracy of dynamic CD by integrating educational prior knowledge, further promoting the development of intelligent education. Martin (1999) explore student models in intelligent tutoring systems, focusing on guiding the teaching process through constraint-based modeling. This approach limits knowledge representation and concentrates on the domain constraints that students either follow or violate when answering questions. **Our FNCD adopts a constraint optimization method for modeling knowledge level in educational teaching.**

Furthermore, Talanquer (2009) analyze the cognitive constraints of students in learning material granularity, revealing the impact of these implicit assumptions on students' thinking and reasoning at different learning stages. In optimization problems, Ungredda and Branke (2024) propose a knowledge gradient acquisition function capable of handling constraints, achieving superior performance in constrained Bayesian optimization. These studies clearly demonstrate that identifying cognitive constraints effectively enhances both the learning experience and the design quality of assessment tools.

Table 1
Symbols and descriptions.

Symbol	Description
S, E	Sets of students and exercises
KC, R	Sets of KCs and responses
n, m, k	Counts of students, exercises, and kcs
Q-matrix	Expert-annotated relationship matrix between exercises and KCs
θ_f	Fine-grained knowledge-level constraint-aware features
θ_{cd}	Diagnostic model-based automatic inference of knowledge mastery features
KPA	Knowledge proficiency assessment via answering performance
BSS	Building student similarity via random grouping
Top-k-AENG	Top-k attention-enhanced graph network
MSRL	Multi-scale relational learning
FNCD	Fine-grained Neural CD with knowledge-level constraint

3. Preliminary

In this section, we elaborate on the fundamental concepts of Fine-Grained Neural Cognitive Diagnosis (FNCD) with Knowledge-Level Constraints and further introduce the key components of the FNCD framework. Table 1 provides a summary of all the symbols used in this paper.

3.1. Problem definition

Under the fine-grained knowledge-level constrained CD framework, we define several core sets: the student set $S = \{s_1, s_2, \ldots, s_n\}$, which contains n students; the exercise set $E = \{e_1, e_2, \ldots, e_m\}$, comprising m exercises; the KC set $KC = \{kc_1, kc_2, \ldots, kc_k\}$, consisting of k KCs; the KC scoring matrix $KS = KS_{ij}^{n\times k}$, where KS_{ij} denotes the average binary score of the ith student on the jth KC; and the response set $R = \{0, 1\}$, recording students' answering behavior, where 1 indicates a completely correct response and 0 represents other cases. These sets serve as the foundation for analyzing students' knowledge mastery and conducting fine-grained assessments.

The students' test records are modeled as a set of triples (s, e, ks), where $s \in S$ represents the student, $e \in E$ denotes the exercise, and $ks \in KS$ is the average binary score of KCs based on prior statistics. To represent the relationship between exercises and KCs, a predefined exercise matrix $Q = Q_{ij}^{m\times k}$, labeled by experts, is used. In this matrix, $Q_{ij} = 1$ if exercise e_i is associated with KC kc_j, and $Q_{ij} = 0$ otherwise. These definitions establish the foundation of the fine-grained knowledge-level constrained CD framework.

Fine-Grained Neural Cognitive Diagnosis (FNCD): Given the students' test records $L = \{(s_i, e_1, ks_i), (s_i, e_2, ks_i), \ldots, (s_i, e_t, ks_i)\}$ and the predefined Q-matrix, our goal is to infer the ability level of students across various KCs. By incorporating fine-grained knowledge-level structures, the mastery of KCs is constrained to ensure the evaluation results align more closely with students' actual ability distributions, reflecting realistic mastery level in educational contexts.

3.2. Basic factors

In the student ability evaluation model, two key vectors are introduced: the diagnostic knowledge mastery vector θ_{cd} and the fine-grained knowledge-level constrained awareness vector $\hat{\theta}_f$, which together reflect a student's proficiency across different KCs.

The diagnostic knowledge mastery vector θ_{cd} is automatically inferred by the model and is computed by multiplying the one-hot encoded student ID $\mathbf{x}_s$ with a trainable knowledge proficiency matrix $\mathbf{S}$:

$$\theta_{cd} = \sigma(\mathbf{x}_s \times \mathbf{S}), \tag{1}$$

where $\theta_{cd} \in (0, 1)^{1\times k}$ represents the student's proficiency across k KCs, $\mathbf{x}_s \in \{0, 1\}^{1\times n}$ is the one-hot encoding of all student IDs, $\mathbf{S} \in \mathbb{R}^{n\times k}$ is the trainable proficiency matrix, and $\sigma(\cdot)$ denotes the sigmoid activation function.

The fine-grained knowledge-level constrained awareness vector θ_f is derived using the students' scoring matrix $\mathbf{ks}_{ij}$ and the knowledge weight vector $\mathbf{k}$:

$$\theta_f = \sigma(\mathbf{ks}_{ij} \times \mathbf{k}), \tag{2}$$

where $\theta_f \in (0, 1)^{1\times k}$ represents the fine-grained knowledge level for a student across all KCs, $\mathbf{ks}_{ij} \in \{0, 1\}^{1\times n}$ is the score encoding, and $\mathbf{k} \in \mathbb{R}^{n\times k}$ is a trainable weight matrix.

In addition, when considering the fundamental attributes of exercises, the model leverages the one-hot encoded exercise ID $\mathbf{x}_e$ to account for the difficulty and discrimination of each item. The Q-matrix, an essential element in CD, captures the relationship between exercises and KCs. Specifically, for an exercise e, its KC relevance vector $\mathbf{Q}_e$ is obtained as follows:

$$\mathbf{Q}_e = \mathbf{x}_e \times \mathbf{Q}, \tag{3}$$

where $\mathbf{x}_e \in \{0, 1\}^{1\times m}$ is the one-hot encoding of exercise IDs, and $\mathbf{Q}_e \in \{0, 1\}^{1\times k}$ represents the KC relevance vector, encapsulating the association between the exercise and its related KCs.

4. The FNCD framework

This section begins with the paradigm of CD, providing a comprehensive introduction to our two-stage, fine-grained neural CD framework with knowledge-level constraint (FNCD). In the following sections, we delve into the two-stage architecture of the proposed framework: Plug-and-Play Fine-Grained Knowledge Modeling and Diagnosis Enhancement, emphasizing the underlying principles and detailing the implementation strategies of each component.

4.1. Model overview

To accurately model students' problem-solving behavior, CDMs typically define a scoring function to estimate their mastery of the underlying knowledge concepts.

$$r = \mathrm{CDM}(\theta, \omega_e), \tag{4}$$

where r represents the students' responses (e.g., scores or correctness), θ denotes their knowledge proficiency, and ω_e encompasses various parameters of the exercise, such as item difficulty $\mathbf{h}_{\mathbf{e}}^{\mathrm{diff}} \in (0, 1)^{1\times k}$ and discrimination $h_e^{\mathrm{disc}} \in (0, 1)$, with k being the number of KCs associated with the exercise.

This modeling process is generally based on CDMs, similar to the response functions used in IRT. The aim is to analyze the relationships between students' performance, their knowledge mastery, and the characteristics of the exercise.

The assessment of students' knowledge level should not solely rely on the model's performance in predicting accuracy, as this approach could lead to learned ability parameters that do not align with the students' true knowledge level. In real educational contexts, parameter values learned by the model in an attempt to improve accuracy may not accurately reflect students' actual abilities, potentially resulting

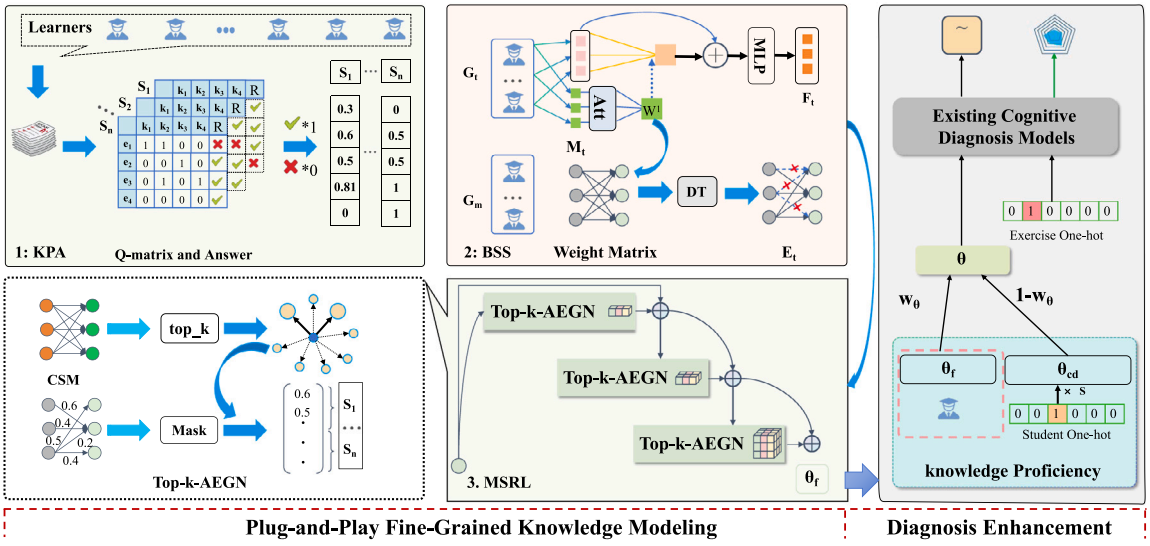


Fig. 2. Overview of the Plug-and-Play FNCD Framework: KPA refers to Knowledge Proficiency Assessment based on students' answering performance. BSS represents Building Student Similarity through random grouping. MSRL stands for Multi-Scale Relational Learning. Top-K-AEGN denotes the Top-K Attention-Enhanced Graph Network. DT refers to a Dynamic Threshold, M_t denotes the input features of the students in group t.

in overestimation or underestimation of their proficiency. Therefore, effective constraints must be imposed on the model's learning process.

To address this, we decompose the students' knowledge proficiency θ into two components to ensure that the ability assessment aligns more closely with their true knowledge level:

$$\theta = \phi(\theta_{cd}, \theta_f), \tag{5}$$

where θ_{cd} represents the knowledge mastery vector inferred by the diagnostic model, and θ_f denotes the ability features constrained by fine-grained knowledge level. The $\phi(\cdot)$ is used to integrate these two types of features, collectively reflecting their overall impact on the students' knowledge proficiency.

Fig. 2 illustrates the overall architecture of the FNCD framework, while Algorithm 1 presents its detailed procedural workflow. The FNCD framework is mainly divided into two stages: the Fine-Grained Knowledge-Level Constraint Perception Modeling stage and the Diagnosis Enhancement stage.

In the Fine-Grained Knowledge-Level Constraint Perception Modeling Stage, we designed three network modules: Knowledge Proficiency Assessment via Answering Performance (KPA), Building Student Similarity via Random Grouping (BSS), and Multi-Scale Relational Learning (MSRL), to simulate the influence of fine-grained knowledge-level constraint perception on knowledge mastery.

In the Diagnosis Enhancement Stage, we integrate the knowledge proficiency from existing CD models with the knowledge proficiency after applying fine-grained knowledge-level constraint perception, using a cognitive behavior function to output predicted scores. After training with student logs, the final knowledge proficiency for each student is generated as the diagnostic result. The details of these two stages are described in the following sections.

4.2. Knowledge proficiency assessment via answering performance (KPA)

To support a more accurate and robust evaluation of students' knowledge proficiency, we introduce a mechanism that leverages students' mastery levels over individual KCs as prior information. This fine-grained assessment enables more effective differentiation among students across varying proficiency levels (Keil, 1981), enhancing the stability and interpretability of the diagnostic process (Martin, 1999). By incorporating KC-level mastery information, the model becomes more resilient to issues such as "overestimation" or "underestimation" of ability. Specifically, we construct prior knowledge representations based on binary scoring of KC performance, which serve as a foundational input for the subsequent proficiency modeling.

Specifically, for each student i and item j, if the student answers correctly (score equals 1), then 1 point is assigned to all KCs involved in that item; otherwise, no points are assigned.

For each student i and knowledge component k, we calculate the accumulated correct count $A_{i,k}$ and total attempt count $C_{i,k}$ on items involving k. The student's proficiency on k is then defined as:

$$\theta_{i,k} = \frac{A_{i,k}}{\max(C_{i,k}, 1)}, \tag{6}$$

where $\max(C_{i,k}, 1)$ prevents division by zero.

At the group level, we aggregate the total correct counts G_k and total attempts H_k across all students:

$$\overline{\theta}_k = \frac{G_k}{\max(H_k, 1)}, \tag{7}$$

which represents the average proficiency on KC k.

To capture individual differences relative to the group, we define the deviation:

$$\Delta\theta_{i,k} = \theta_{i,k} - \overline{\theta}_k. \tag{8}$$

4.3. Building student similarity via random grouping (BSS)

To better model collaborative learning patterns among students and capture potential group-level relationships, we design a student similarity extraction module that constructs local similarity graphs by randomly grouping students. This module aims to extract intra-group interaction patterns, reflecting the learning dynamics within small groups while reducing noise interference and computational complexity in global modeling.

Specifically, we randomly divide the student set S into m disjoint subsets $G = \{G_1, G_2, \ldots, G_m\}$. This random grouping strategy ensures both simplicity and computational efficiency by effectively leveraging local aggregation patterns in students' knowledge state, while avoiding the bias and instability introduced by score- or performance-based grouping. Jenkins et al. (2023) have shown that fair and diverse grouping strategies can enhance learning outcomes and improve model robustness and generalization.

Algorithm 1 FNCD Framework

Input:
Student response record matrix $R \in \{0,1\}^{n\times m}$
Q-matrix $Q \in \{0,1\}^{m\times K}$
Number of groups m
Multi-scale propagation depth L
Output: Student knowledge proficiency representation θ and prediction result r
Stage One: Fine-grained Knowledge Constraint Perception Modeling
Calculate the mastery level of each student on every knowledge concept θ_f according to Eqs. (6)–(8);
Randomly divide the student set S into m groups;
for each group t **do**
Use Eq. (9) to compute intra-group features $\mathbf{A}_t$ and similarity matrix $\mathbf{W}_t$ via multi-head attention mechanism;
Use Eqs. (10)–(11) to calculate threshold τ_t and construct sparse intra-group edge set $\mathbf{E}_t^{\text{intra}}$;
Obtain intra-group student features $\mathbf{F}_t^{\text{intra}}$ using a three-layer perceptron network as in Eq. (12);
Calculate the group-level representation $\mathbf{g}_t$ by averaging group features according to Eq. (13);
end for
Based on group-level features, compute inter-group cosine similarity S_{pq} and select Top-k neighbors $\mathcal{N}_p^{(0)}$ using Eqs. (14)–(15);
Construct sparse adjacency matrix $\mathbf{A}_{\text{inter}}^{(0)}$ by Eq. (16);
for $l = 0$ to $L-1$ **do**
Obtain neighbor set $\mathcal{N}_p^{(l)}$ using Eq. (17);
Compute attention weights $\alpha_{pq}^{(l)}$ as in Eq. (18);
Aggregate neighbor features and update node representation $\mathbf{g}_p^{(l+1)}$ according to Eq. (19);
end for
Fuse all scale representations and obtain the final group-level representation $\mathbf{g}_p^{\text{inter}}$ via Eq. (20);
Stage Two: Diagnosis Enhancement Stage
Fuse θ_{cd} and θ_f with personalized weights to get the final knowledge proficiency θ according to Eq. (21);
Predict student scores r using the cognitive diagnosis model function as in Eq. (22);
Optimize model parameters by minimizing the cross-entropy loss function, as shown in Eq. (23);
return θ, r

Within each group G_t, we employ a multi-head attention mechanism $\mathcal{F}_{\text{attn}}(\cdot)$ to extract potential interaction features among students, obtaining the transformed feature matrix $\mathbf{A}_t$ and the intra-group similarity matrix $\mathbf{W}_t$:

$$\mathbf{A}_t, \mathbf{W}_t = \mathcal{F}_{\text{attn}}(\mathbf{M}_t), \tag{9}$$

where $\mathbf{M}_t$ denotes the input features of students in the tth group.

To filter out insignificant relationships and suppress weak or noisy connections, we adaptively determine a threshold based on the similarity distribution:

$$\tau_t = \mu_t + \alpha \cdot \sigma_t, \tag{10}$$

where μ_t and σ_t denote the mean and standard deviation of the similarity matrix $\mathbf{W}_t$, respectively. In this work, the hyperparameter α is set to 0.5, consistent with empirical insights from prior cognitive diagnosis research (e.g., HCD (Sun et al., 2025a) and LGS-KT (Sun et al., 2025c)), where similar values have been shown to balance graph sparsity and semantic retention effectively. We retain only those edges with similarity above the threshold to construct a sparse but semantically meaningful edge set:

$$\mathbf{E}_t^{\text{intra}} = \{(i,j) \mid \mathbf{W}_{t,ij} > \tau_t\}, \tag{11}$$

yielding the intra-group student connection structure $\mathbf{E}_t^{\text{intra}}$, which captures strong relational patterns.

Finally, we feed the feature matrix $\mathbf{A}_t$ into a multi-layer perceptron (MLP) to generate compact, nonlinearly transformed intra-group relational representations:

$$\mathbf{F}_t^{\text{intra}} = f_{\text{MLP}}(\mathbf{A}_t; \boldsymbol{\Theta}), \tag{12}$$

where $f_{\text{MLP}}(\cdot)$ denotes a three-layer perceptron with ReLU activation.

4.4. Multi-scale relational learning (MSRL)

To enhance the model's ability to capture students' knowledge state across groups, this module aims to dynamically construct the relational structure among students and implement a multi-scale feature interaction mechanism. This enhances the expressive power of student features and improves the accuracy of cognitive state representations. Unlike relying on static prior structures, this module adaptively builds graph structures based on students' knowledge proficiency features and introduces multi-scale interaction mechanisms to capture potential collaborative learning relationships and semantic structures at multiple hierarchical levels from local to global, thereby improving the generalization and modeling depth of representations.

4.4.1. Group-level student feature construction

First, based on the student knowledge proficiency feature representations $\mathbf{F}_t^{\text{intra}}$ obtained from the previously described BSS module, we aggregate the features of student nodes within each group G_t to obtain the group-level representation $\mathbf{g}_t$, defined as:

$$\mathbf{g}_t = \frac{1}{|G|} \sum_{t\in G} \mathbf{F}_t^{\text{intra}}, \tag{13}$$

where $\mathbf{g}_t$ denotes the average feature representation of group G_t, characterizing the overall knowledge state of the students in the group.

4.4.2. Inter-group neighbor construction

Based on the group-level feature representations $\{\mathbf{g}_t\}_{t=1}^m$, the cosine similarity between any two groups is computed as:

$$S_{pq} = \frac{\mathbf{g}_p \cdot \mathbf{g}_q}{\|\mathbf{g}_p\| \cdot \|\mathbf{g}_q\|}, \quad p,q \in \{1,\ldots,m\}. \tag{14}$$

Subsequently, for each group p, the top k most similar neighboring groups are retained to form the initial neighbor set, balancing graph sparsity and information richness while reducing noise and overfitting (Gao et al., 2024).

$$\mathcal{N}_p^{(0)} = \text{TopK}(\{S_{pq} \mid q \neq p\}, k), \tag{15}$$

where k is fixed as 3 in this study.

Based on this neighbor set, a sparse adjacency matrix $\mathbf{A}_{\text{inter}}^{(0)} \in \mathbb{R}^{m\times m}$ is constructed, with entries defined by:

$$\mathbf{A}_{\text{inter}}^{(l)}[p,q] = \begin{cases} S_{pq}, & q \in \mathcal{N}_p^{(0)} \\ 0, & \text{otherwise.} \end{cases} \tag{16}$$

4.4.3. Multi-scale inter-group interaction learning

followed by sparsification and normalization to retain the most important neighbors, yielding the neighbor set at scale l:

$$\mathcal{N}_p^{(l)} = \{q \mid \mathbf{A}_{\text{inter}}^{(l)}[p,q] > 0\}. \tag{17}$$

To account for the varying importance of different neighbors to the current group, an attention-based message passing strategy is adopted

at the lth scale graph layer to compute the attention weight for group p and its neighbor $q \in \mathcal{N}_p^{(l)}$:

$$\alpha_{pq}^{(l)} = \frac{\exp\left(\phi\left(\mathbf{a}^\top[\mathbf{g}_p^{(l)} \oplus \mathbf{g}_q^{(l)}]\right)\right) \cdot \mathbf{A}_{\text{inter}}^{(l)}[p,q]}{\sum_{r \in \mathcal{N}_p^{(l)}} \exp\left(\phi\left(\mathbf{a}^\top[\mathbf{g}_p^{(l)} \oplus \mathbf{g}_r^{(l)}]\right)\right) \cdot \mathbf{A}_{\text{inter}}^{(l)}[p,r]}, \tag{18}$$

where $\oplus$ denotes feature concatenation, $\phi(\cdot)$ is the LeakyReLU activation function, and $\mathbf{a}$ is a learnable attention parameter vector.

The node representation is updated at each layer by aggregating the weighted neighbor features:

$$\mathbf{g}_p^{(l+1)} = f\left(\mathbf{g}_p^{(l)} + \sum_{q \in \mathcal{N}_p^{(l)}} \alpha_{pq}^{(l)} \mathbf{g}_q^{(l)}; \Theta_l\right), \tag{19}$$

where $f(\cdot)$ denotes a nonlinear transformation function, and Θ_l represents the learnable parameters at layer l.

After L layers of interaction, the multi-scale representations are fused to obtain the final group-level feature:

$$\mathbf{g}_p^{\text{inter}} = f\left(\mathbf{g}_p^{(0)} + \sum_{l=0}^{L-1} \sum_{q \in \mathcal{N}_p^{(l)}} \alpha_{pq}^{(l)} \mathbf{g}_q^{(l)}; \Theta\right), \tag{20}$$

where $\Theta = \{\Theta_0, \ldots, \Theta_{L-1}\}$ denotes the set of parameters across all layers. In this paper, l is set to 3 to prioritize computational efficiency and practical applicability, rather than pursuing ultimate accuracy.

4.5. Diagnosis enhancement

At this stage, the goal is to construct a comprehensive representation of each student's knowledge state. This is achieved by fusing two complementary sources of information: (i) fine-grained, constraint-aware features that reflect knowledge-level dependencies, and (ii) knowledge mastery representations inferred by the diagnostic model. To effectively integrate these heterogeneous features, we introduce a personalized adaptive weighting mechanism, which dynamically adjusts the fusion process based on individual learning characteristics. This design ensures that the resulting representation more accurately captures the unique knowledge mastery trajectory of each student.

$$\theta = w_{\theta_i} \cdot \theta_{cd} + (1 - w_{\theta_i}) \cdot \theta_f, \tag{21}$$

where w_{θ_i} is the personalized weighting coefficient for student i, optimized during model training based on the student's learning characteristics.

Building on this, the model employs the core principles of CD to predict student performance during practice. The prediction is based on the following formula: $r = \mathbf{Q}_e \circ \left(\theta - \mathbf{h}_{\mathbf{e}}^{\text{diff}}\right) \times h_e^{\text{disc}}$, a method applicable to various cognitive diagnostic models, such as IRT, MIRT, and NCDM.

$$r = \text{CDM}(\theta, \omega_e), \tag{22}$$

where ω_e includes exercise-related parameters such as difficulty ($\mathbf{h}_{\mathbf{e}}^{\text{diff}}$) and discrimination ($h_e^{\text{disc}}$).

By effectively integrating knowledge representations driven by CD with fine-grained learning constraints, this approach accounts for individual differences among students, enabling more accurate predictions of student abilities and dynamic adjustments to their learning paths.

It is worth noting that the computational complexity of NCDM is $\mathcal{O}(L \cdot d^2)$, while the FNCD framework introduces an additional sparse graph attention module with a complexity of $\mathcal{O}(H \cdot E \cdot d)$, where L denotes the sequence length, d is the feature dimension, E is the number of edges in the graph (with $E \ll L^2$ due to the sparse graph construction), and H represents the number of scales for fusion. In this work, a sparse graph structure is constructed and three scales are used for relation modeling. Compared to NCDM, the FNCD framework effectively enhances cognitive diagnostic performance while significantly controlling the increase in model complexity through local graph construction, sparse connectivity, and multi-scale fusion strategies, demonstrating favorable computational efficiency and scalability.

4.6. Objective function

To optimize all parameters within the FNCD framework, we incrementally analyze each response log record in the test data and use the cross-entropy loss function to measure the discrepancy between the predicted value $\tilde{y}$ and the actual student response label r. The specific form of the loss function is as follows:

$$\mathcal{L}(\tilde{y}, r) = -\sum_i \left(r_i \log(\tilde{y}_i) + (1 - r_i)\log(1 - \tilde{y}_i)\right), \tag{23}$$

to minimize this loss function, the Adam optimization algorithm is employed.

5. Experiments

In this section, we introduce the real-world dataset used for the experiments, provide a detailed explanation of key aspects of the training process, and describe the baseline models selected for comparison. Subsequently, we evaluate the impact of integrating the fine-grained knowledge-level constraint features proposed in this study into the baseline models on student performance prediction. To further elucidate the effect of FNCD on model performance and knowledge state assessment, we delve into the following research questions:

- RQ1: How does the proposed FNCD framework improve and influence student performance prediction compared to existing frameworks?
- RQ2: What is the performance of each module within the FNCD framework?
- RQ3: Can the FNCD framework achieve fine-grained constraints on student knowledge state, thereby enhancing the rationality of predictions?
- RQ4: What impact does the FNCD framework have on the process of modeling student knowledge state?
- RQ5: How does the diagnostic output of the FNCD framework perform in terms of interpretability?
- RQ6: What impact do the modeling design factors have on the performance of FNCD-NCDM?
- RQ7: What role do the fine-grained knowledge-level constraint features and the diagnostic model's automated knowledge mastery representations play in improving student cognition?
- RQ8: How does the FNCD framework enhance the diagnostic outcomes for students, and in what specific aspects is this improvement reflected?

5.1. Datasets and experimental setup

We validate the effectiveness of the fine-grained knowledge-level constraint CD system using data from the 2015 Programme for International Student Assessment (PISA 2015)[4]. This dataset includes over 500,000 students from 73 countries and regions, focusing on the subjects of science, reading, and mathematics. During the data preprocessing stage, items marked as "Full credit" were considered correct (labeled as 1), while "Partial credit" and "No credit" were treated as incorrect (labeled as 0); unanswered items were marked as missing responses. To ensure the accuracy of the analysis, data from students with fewer than 30 responses were excluded, and for the mathematics section, records with fewer than 20 responses were removed.

The Junyi dataset was collected from Junyi Academy, a Chinese online learning platform dedicated to mathematics education. It comprises student interaction logs recorded from October 2012 to January 2015, along with rich metadata about the exercises and their interrelations. In

[4] https://www.oecd.org/en/about/programmes/pisa/pisa-data.html

Table 2
Statistics of all datasets.

Statistics \ Datasets	Science	Read	Math	Junyi
# of exercise records	7,585,349	811,136	2,594,902	353,835
# of correct answers	3,691,123	528,570	1,206,973	212,749
# of students	233,405	26,881	116,805	10,000
# of exercises	184	103	81	835
# of KCs	21	10	11	835
Average of exercise	32.50	30.18	22.22	35.38

the preprocessing phase, we retained only each student's first attempt per exercise and filtered out students who had attempted fewer than 15 questions to ensure statistical reliability. After processing, the dataset was randomly divided into training (70%), validation (10%), and test (20%) sets, as detailed in Table 2. At the beginning of the experiments, we initialized the model parameters using the Xavier method and trained the model with five-fold cross-validation. The final result was obtained by averaging the performance across all folds. During optimization, the initial learning rate was set to 0.002, and the dropout rate was 0.5, which is a common setting in neural cognitive diagnosis models. The mini-batch size was set to 32, and no additional weight decay was applied during training. All hyperparameters were set based on the training set, and the model with the best performance on the validation set was selected for evaluation on the test set. The experiments were conducted on a server configured with a 64-bit Ubuntu 20.04.5 LTS operating system, equipped with a 2.30 GHz Intel Xeon Gold 5218 CPU and a 32 GB Tesla V100 GPU. The computations were implemented using the PyTorch framework.

5.2. Baselines

To investigate the impact of fine-grained knowledge-level constraints on model performance, we developed eight models based on the FNCD framework: FNCD-IRT, FNCD-MIRT, FNCD-DINA, FNCD-NCDM, FNCD-AKT, FNCD-RCD, FNCD-TKT-CD and FNCD-MHCD. These models integrate classical CD methods. The models were reproduced[5] and carefully fine-tuned to ensure each operates at its optimal state, ensuring a fair comparison. We used Area Under the Curve (AUC), Accuracy (ACC), and Root Mean Square Error (RMSE) as the primary performance metrics to comprehensively evaluate the models.

- **IRT**(Lord, 2012): This model assesses students' performance on different test items to measure their mastery of specific KCs, estimating both student ability and item difficulty.
- **MIRT**(Ackerman, 2014): An extension of IRT, the MIRT model analyzes students' performance across multiple ability dimensions, estimating multiple ability parameters and item characteristics simultaneously.
- **DINA**(De La Torre, 2009): This model assumes each item involves multiple KCs and uses students' responses to infer their mastery of these concepts, employing a binary classification approach for representation.
- **NCDM**(Wang et al., 2022): Leveraging neural networks, the NCDM model captures the complex relationships between students and test items, offering accurate and interpretable CD as an alternative to traditional handcrafted predictive functions.
- **HCD**(Sun et al., 2025a): The HCD model integrates students' prior statistical information to construct hierarchical structures and introduces hierarchy-aware neural networks as a flexible, plug-and-play solution. This framework deeply explores complex relationships across different hierarchical level of students, enabling more accurate modeling of students' knowledge state in real-world educational contexts.
- **AKT**(Ghosh et al., 2020): The AKT is an interpretable attention-based model that enhances knowledge tracing by incorporating cognitive principles and a monotonic attention mechanism to improve both prediction accuracy and personalization.
- **RCD**(Gao et al., 2021) The RCD model achieves more accurate student cognitive diagnosis by constructing a multi-layer relation map that includes student-exercise interactions, concept-exercise correlations, and concept dependencies, and by using a multi-level attention network to uniformly model these relationships.
- **TKT-CD**(Sun et al., 2025d) The TKT-CD model introduces two core modules, time-driven and cognition-driven, to dynamically track the transitions of a student's knowledge state during exercises and exams, in order to overcome the limitations of traditional static diagnosis models and enhance diagnostic accuracy.
- **MHCD**(Sun et al., 2025f): The MHCD is an extension of HCD that enhances knowledge state modeling by introducing multi-hierarchy feature interactions and iterative constraint-aware inference for improved accuracy and interpretability.

5.3. Performance prediction (RQ1)

Since students' true knowledge state are challenging to measure directly, evaluating the performance of CD models remains a significant challenge. Although improving predictive accuracy is not the primary goal of our study, analyzing model performance in student performance prediction tasks provides an indirect validation of the model's rationale. This approach also highlights the core value of FNCD in enhancing CD effectiveness.

Table 3 presents the performance evaluation results of CD models integrated with FNCD. The analysis reveals that FNCD-based models (e.g., FNCD-IRT) significantly outperform traditional models (e.g., IRT) across all metrics, demonstrating the critical role of fine-grained knowledge constraints in CD. Moreover, integrating FNCD into the NCDM and MHCD model achieved the best performance across AUC, ACC, and RMSE metrics, highlighting the effectiveness of neural networks in capturing students' knowledge state features. Although the PISA dataset is designed for static cognitive diagnosis, which limits the effectiveness of knowledge tracing models, incorporating our FNCD into the knowledge tracing model AKT still resulted in notable performance improvements. Furthermore, FNCD also demonstrates a significant performance improvement effect on more complex neural network models, as evidenced by models like FNCD-RCD and FNCD-TKT-CD.

A comparison of the four datasets further shows that FNCD exhibits strong adaptability across varying data scales. Whether applied to Science, a dataset with abundant data, or Read, which has a smaller dataset, FNCD achieved satisfactory results. Even in Math, where interactive data is sparse, the framework demonstrated remarkable performance. This suggests that, compared to HCD, which focuses on overarching hierarchical constraints for students, FNCD provides a more precise reflection of students' knowledge mastery by refining constraints to the knowledge level.

5.4. Ablation study (RQ2)

To analyze the roles of the three core modules in the FNCD framework – MSRL, BSS, and KPA – we designed ablation experiments. Specifically, "−MSRL" denotes the removal of the MSRL module while retaining the other components of the framework. "−BSS" represents the replacement of the BSS module with randomly generated intra-group similarity edges, while maintaining the other structural elements. "−KPA" involves substituting the KPA module with randomly initialized prior knowledge proficiency values for students, while the other components remain unchanged.

The experimental results are presented in Table 4, and a detailed analysis follows. First, the removal of any single module leads to

[5] https://github.com/bigdata-ustc/EduCDM

Table 3
Comparison of student performance prediction across different methods. * denotes a p-value less than 0.05 in t-test.

Methods	Science			Read			Math			Junyi		
	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE
IRT	0.622	0.615	0.559	0.638	0.623	0.544	0.643	0.622	0.542	0.775	0.676	0.427
HCD-IRT	0.651	0.633	0.536	0.664	0.654	0.529	0.675	0.665	0.524	0.803	0.715	0.403
FNCD-IRT	**0.705***	**0.695***	**0.513***	**0.714***	**0.701***	**0.511***	**0.726***	**0.714***	**0.495***	0.813*	0.726*	0.401*
MIRT	0.665	0.635	0.524	0.678	0.647	0.519	0.668	0.642	0.516	0.799	0.751	0.412
HCD-MIRT	0.698	0.675	0.507	0.709	0.686	0.504	0.694	0.684	0.502	0.833	0.793	0.397
FNCD-MIRT	**0.734***	**0.721***	**0.487***	**0.719***	**0.708***	**0.475***	**0.752***	**0.739***	**0.425***	0.846*	0.813*	0.389*
DINA	0.677	0.648	0.547	0.689	0.657	0.539	0.682	0.655	0.533	0.735	0.679	0.463
HCD-DINA	0.723	0.703	0.535	0.736	0.717	0.533	0.729	0.708	0.524	0.769	0.713	0.424
FNCD-DINA	**0.785***	**0.751***	**0.421***	**0.794***	**0.776***	**0.411***	**0.809***	**0.784***	**0.409***	0.775*	0.724*	0.421*
NCDM	0.789	0.712	0.434	0.792	0.744	0.417	0.806	0.727	0.424	0.761	0.714	0.445
HCD-NCDM	0.812	0.732	0.422	0.822	0.762	0.401	0.837	0.755	0.411	0.798	0.738	0.413
FNCD-NCDM	**0.843***	**0.757***	**0.402***	**0.845***	**0.782***	**0.388***	**0.874***	**0.791***	**0.381***	0.813*	0.743*	0.402*
AKT	0.629	0.614	0.542	0.647	0.624	0.495	0.649	0.623	0.486	0.829	0.807	0.403
HCD-AKT	0.634	0.629	0.531	0.651	0.632	0.487	0.658	0.637	0.476	0.834	0.813	0.392
FNCD-AKT	**0.652***	**0.645***	**0.501***	**0.672***	**0.644***	**0.471***	**0.677***	**0.641***	**0.470***	**0.854***	**0.833***	**0.379***
RCD	0.785	0.705	0.440	0.790	0.735	0.420	0.801	0.720	0.430	0.755	0.710	0.450
HCD-RCD	0.805	0.725	0.431	0.815	0.755	0.412	0.828	0.745	0.419	0.790	0.731	0.423
FNCD-RCD	0.835*	0.751*	0.411*	0.839*	0.735*	0.395*	0.865*	0.782*	0.390*	0.805*	0.740*	0.412*
TKT-CD	0.787	0.710	0.438	0.795	0.740	0.415	0.804	0.725	0.428	0.763	0.712	0.448
HCD-TKT-CD	0.810	0.730	0.425	0.819	0.760	0.405	0.835	0.751	0.415	0.795	0.735	0.418
FNCD-TKT-CD	0.841*	0.755*	0.405*	0.842*	0.780*	0.391*	0.871*	0.788*	0.385*	0.810*	0.741*	0.405*
MHCD	0.822	0.743	0.402	0.835	0.783	0.401	0.847	0.766	**0.406**	0.859	0.799	0.381
FNCD-MHCD	**0.855***	**0.769***	**0.384***	**0.860***	**0.794***	**0.391***	**0.889***	**0.813***	**0.376***	**0.867***	**0.803***	**0.379***

Table 4
Ablation study of the FNCD module based on the classic and fundamental diagnostic models IRT, MIRT, DINA, and NCDM.

Methods	Science			Read			Math			Junyi		
	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE	AUC	ACC	RMSE
FNCD-IRT	**0.705**	**0.695**	**0.513**	**0.714**	**0.701**	**0.511**	**0.726**	**0.714**	**0.495**	**0.813**	**0.726**	**0.401**
- MSRL	0.681	0.671	0.538	0.681	0.680	0.531	0.705	0.691	0.501	0.784	0.702	0.425
- BSS	0.678	0.663	0.540	0.673	0.669	0.533	0.684	0.672	0.506	0.779	0.698	0.431
- KPA	0.635	0.629	0.548	0.648	0.628	0.540	0.645	0.627	0.537	0.755	0.679	0.443
FNCD-MIRT	**0.734**	**0.721**	**0.487**	**0.719**	**0.708**	**0.475**	**0.752**	**0.739**	**0.425**	**0.846**	**0.813**	**0.389**
- MSRL	0.716	0.697	0.514	0.691	0.673	0.515	0.728	0.714	0.512	0.820	0.784	0.408
- BSS	0.709	0.691	0.516	0.686	0.665	0.517	0.717	0.700	0.513	0.812	0.779	0.415
- KPA	0.674	0.641	0.522	0.689	0.656	0.518	0.684	0.667	0.515	0.791	0.754	0.423
FNCD-DINA	**0.785**	**0.751**	**0.421**	**0.794**	**0.776**	**0.411**	**0.809**	**0.784**	**0.409**	**0.775**	**0.724**	**0.421**
- MSRL	0.765	0.731	0.507	0.769	0.749	0.573	0.776	0.754	0.506	0.753	0.701	0.441
- BSS	0.756	0.729	0.511	0.760	0.747	0.575	0.762	0.748	0.510	0.746	0.695	0.448
- KPA	0.723	0.712	0.520	0.731	0.701	0.577	0.733	0.702	0.523	0.721	0.678	0.455
FNCD-NCDM	**0.843**	**0.757**	**0.402**	**0.845**	**0.782**	**0.388**	**0.874**	**0.791**	**0.381**	**0.813**	**0.743**	**0.402**
- MSRL	0.826	0.739	0.427	0.818	0.765	0.414	0.841	0.763	0.425	0.790	0.721	0.425
- BSS	0.818	0.732	0.429	0.810	0.756	0.417	0.828	0.751	0.426	0.780	0.715	0.431
- KPA	0.787	0.721	0.438	0.796	0.749	0.420	0.795	0.734	0.437	0.756	0.700	0.442

performance degradation, indicating that each module is essential for the overall effectiveness of the framework and validating its role in modeling relationships at various level. Second, given that the dataset lacks inherent edge relationships, the appropriate construction and efficient learning of these edges have a significant impact on model performance. Finally, the removal of the KPA module results in the most substantial performance decline, demonstrating that prior statistics-based student knowledge state are crucial for constraining the learning process, particularly when modeling fine-grained knowledge level.

5.5. Performance of knowledge-level constraint diagnosis (RQ3)

The introduction of fine-grained knowledge level constraints aims to enhance the precision with which existing CD models characterize students' relative ability performance, enabling more accurate inferences of students' proficiency in various KCs. This constraint is embedded in the model through prior knowledge, ensuring that the assessment results align more closely with real-world scenarios. Fig. 3 presents the application effect of fine-grained knowledge level constraints on the Math dataset in the form of kernel density plots.

Several important conclusions can be drawn from the figure. First, both the original NCDM model and the FNCD-NCDM model, which incorporates fine-grained knowledge level constraints, are capable of distinguishing differences in ability distributions among students for the same KC. Second, comparing Figure a and Figure c reveals that the FNCD-NCDM model maintains a high degree of consistency with prior statistics in fitting the distribution of a single KC, demonstrating that the proposed fine-grained knowledge level constraint plays a crucial role in guiding the model to learn reasonable distributions. In contrast, the NCDM model, lacking additional constraints and relying solely on objective function optimization, exhibits a significant deviation from the prior distribution. Overall, the results indicate that the FNCD-NCDM model provides a more accurate assessment of students' knowledge state.

Further analysis of the distribution density for students with ability values close to 0.5 can more clearly validate the role of the fine-grained knowledge level constraint. For example, for KC_{11}, the prior

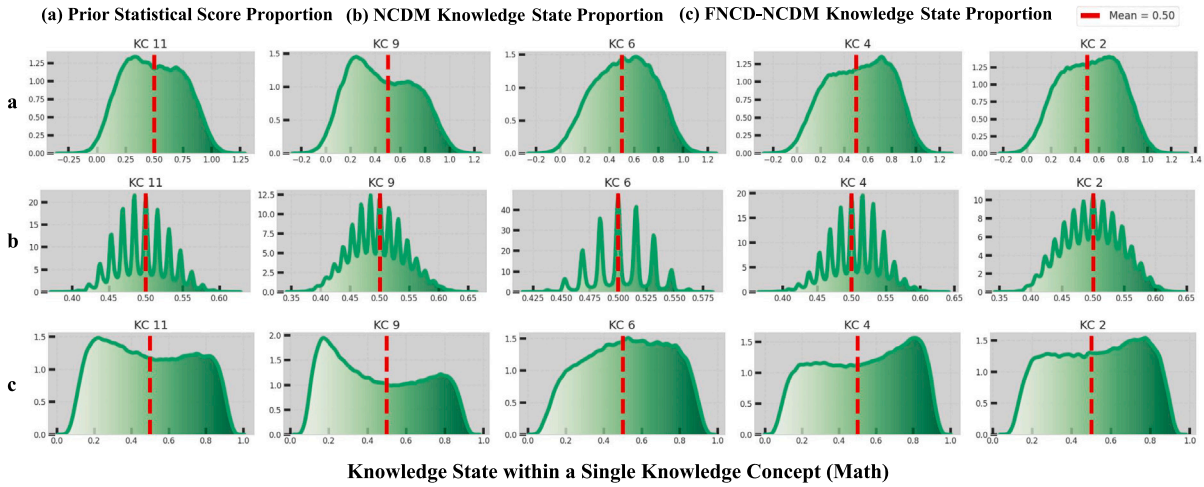


Fig. 3. Comparison of knowledge state learning for 5 randomly selected KCs from the Math dataset, focusing on prior statistical knowledge state, NCDM, and FNCD-NCDM.

distribution shows that approximately 1.1% of students have an ability value of 0.5, while the NCDM model's result is as high as 18.2%. In comparison, the FNCD-NCDM model shows an occurrence rate of 1.2% at this stage, which aligns more closely with the prior distribution. This result indicates that the fine-grained knowledge level constraint significantly improves the model's ability to infer students' knowledge state.

5.6. Knowledge state modeling (RQ4)

The Fig. 3 illustrates the overall effect of incorporating fine-grained knowledge-level constraints under group effects in the Science dataset, while the specific impact of these constraints at the individual level remains to be further explored. The students' knowledge mastery vectors were computed using both the prior statistical KPA method and the FNCD framework proposed in this paper. We focused on analyzing the comparison results between the target student and the top three most similar students. Fig. 4 summarizes the key observations.

First, both KPA and FNCD methods can identify students similar to the target student within the group. Second, we observed that the selection of similar students did not change significantly between the two methods in most cases. For example, in Group 3, both KPA and FNCD identified the same similar students. However, in other groups, there were certain discrepancies in the choice of similar students. For instance, in Group 1, based on the KPA method, student s_1^1 was similar to student s_2^1, whereas under the FNCD method, the similar student was changed to s_9^1. Similarly, in Group 2, the similar student for s_1^2 shifted from s_4^2 in the KPA method to s_{25}^2 in the FNCD method.

Although the similarity results were not entirely consistent, these findings suggest that fine-grained knowledge level constraints contribute positively to individual knowledge state assessment, thereby validating their practical value in cognitive diagnosis.

5.7. Interpretability of diagnostic results (RQ5)

In CD, a student's mastery level of KCs significantly impacts their performance on tests, with higher mastery students having a higher success rate (Chen et al., 2017). To assess the interpretability of the model, the Degree of Agreement (DOA) metric was introduced (Fouss et al., 2007).

Traditional models (e.g., IRT and MIRT) describe student abilities using latent trait vectors but struggle to refine these abilities to specific KCs. This study compares FNCD's DOA performance with baseline models (Random, DINA, NCDM), highlighting differences in how models capture student knowledge state (As shown in 5).

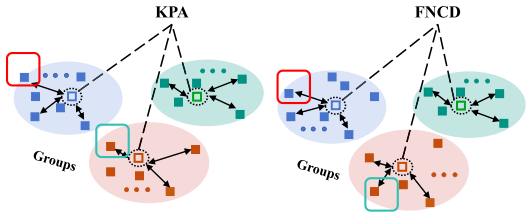


Fig. 4. Randomly select 3 groups from the Science dataset for comparison of fine-grained knowledge level constraints.

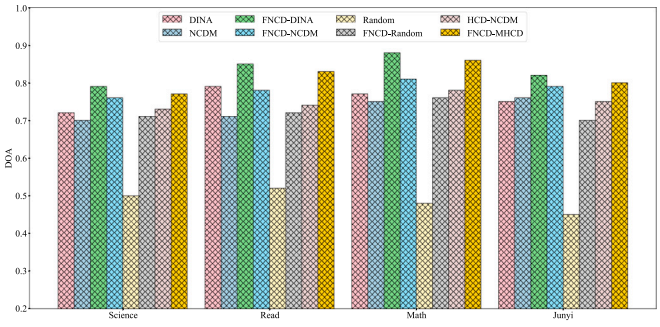


Fig. 5. Comparative analysis of model DOA interpretability across three distinct datasets.

The results show that although the DINA model lacks constraint perception, its DOA values are outstanding across all datasets, indicating the theoretical framework's advantage in interpretability. However, the model's predictive performance is limited. In contrast, models incorporating fine-grained knowledge constraints show higher DOA values, significantly enhancing interpretability. Furthermore, the effect of fine-grained constraints is superior to hierarchical constraints (e.g., HCD-NCDM), suggesting that detailed analysis of KCs is more valuable.

It is noteworthy that the DOA of the Random model remains stable around 0.5, while the FNCD-Random model with fine-grained knowledge constraints significantly improves, approaching that of NCDM and HCD-NCDM. This further highlights the importance of fine-grained knowledge perception in enhancing interpretability.

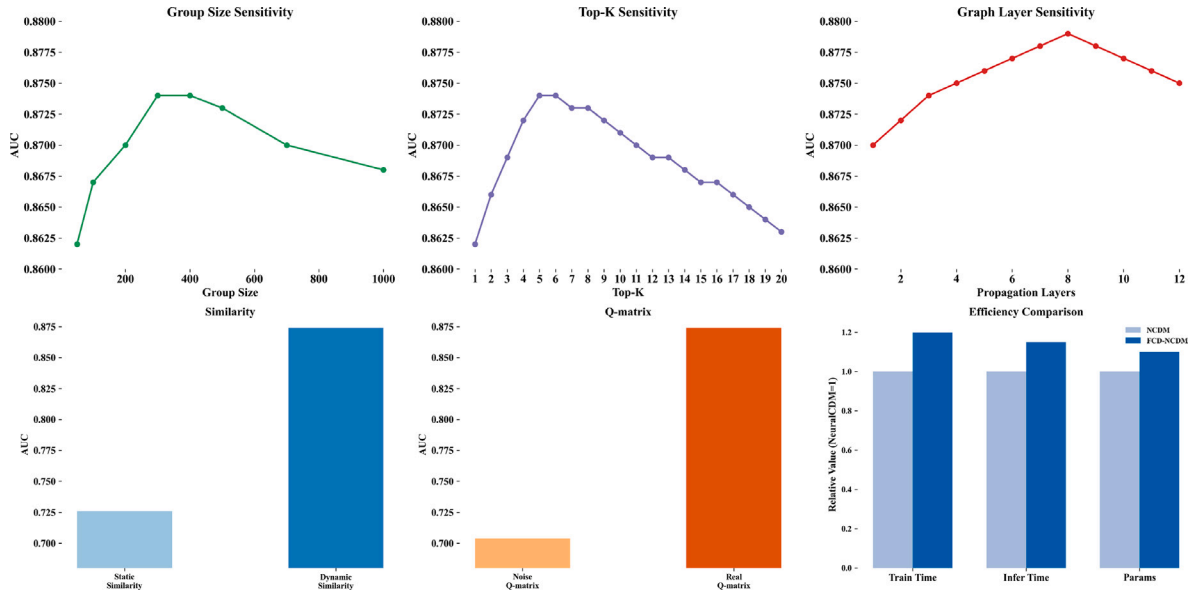


Fig. 6. Impact of similarity modeling, group size, graph layer, and Top-K on FNCD-NCDM performance.

5.8. Mechanism analysis (RQ6)

To systematically evaluate the contributions of each module in FNCD-NCDM, we designed multiple comparative experiments, with results shown in Fig. 6. First, in the sensitivity analysis of group size, model performance steadily improved as the group size increased from 50 to 300, reaching a peak AUC of 0.874. Performance slightly declined when the group size became too large, indicating that moderate aggregation helps to capture latent relations among students, while excessively large group sizes may introduce redundant information. Second, the sensitivity analysis on the number of Top-K neighbors demonstrated that the optimal performance (AUC = 0.874) was achieved when Top-K was set to 5. Increasing the number of neighbors beyond this slightly degraded performance, suggesting that an appropriate number of neighbors benefits the quality of similarity modeling. It should be noted that since this paper does not take pursuing accuracy as the primary evaluation indicator, but rather focuses more on the overall efficiency of the model in practical applications, the value of Top-K is set to 3 after comprehensive consideration.

Third, the sensitivity analysis on the number of propagation layers showed that model performance continuously improved as the layers increased from 1 to 8, reaching a peak AUC of 0.879. Further increasing the layers resulted in stable or slightly decreased performance, reflecting that an appropriate depth facilitates information propagation and effective constraints on students' knowledge state. Subsequently, in the comparison experiments on dynamic versus static similarity and Q-matrix quality, the introduction of dynamic similarity improved the AUC from 0.726 to 0.874. Replacing the noisy Q-matrix with the true Q-matrix also increased the AUC from 0.704 to 0.874, verifying the importance of relationship modeling and high-quality prior information for accurate reasoning. Finally, the efficiency comparison showed that, compared to the original NCDM, FNCD-NCDM increased training time, inference time, and parameter count by approximately 20%, 15%, and 10%, respectively, while delivering significant performance gains, demonstrating a favorable balance between effectiveness and efficiency. It is important to emphasize that this work does not blindly pursue the highest AUC but aims to enhance performance while maintaining low computational cost and effective constraints on students' knowledge state, ensuring the model's practicality and interpretability for real-world applications.

5.9. Visualized diagnosis enhancement (RQ7)

Fig. 7 illustrates the synergistic effect and influence of the fine-grained knowledge level constraint ability θ_f and the diagnostic model's automatically inferred knowledge mastery vector θ_{cd} on student abilities. From the figure, it is evident that θ_{cd} and θ_f exhibit significant differences across different students, highlighting the importance of separately considering these two types of abilities within the model. Further analysis reveals that the contributions of θ_{cd} and θ_f to student ability are not the same. For example, student S_1 has a higher θ_{cd} value compared to θ_f, but θ_f has a more significant impact on their final ability θ; conversely, student S_{23} scores higher in θ_f but is more influenced by θ_{cd}. This indicates that the abilities of different students are affected by these two features in distinct ways. Finally, the regression line for the fine-grained knowledge level ability is higher than that for the diagnostic model's automatically inferred knowledge mastery representation, suggesting that the fine-grained knowledge level ability has a more substantial impact on the final ability in the performance of these 30 students.

5.10. Case study (RQ8)

The relationship between students' CD results, their ability, and the difficulty of the questions can be quantitatively analyzed using Eq. (22). We randomly selected two students from the Science and Math datasets, along with six questions they answered. The upper part of Fig. 8(a) shows the diagnostic results of two students from the Science dataset, while the lower part (b) displays the diagnostic results of two students from the Math dataset.

From the figure, it is evident that when students' knowledge mastery and the requirements of the questions are clear, the model's predictions are highly consistent with students' actual responses. For example, in the case of student S_1 shown in Figure (a), the comparison between the student's final ability value θ and the question difficulty shows that only for question e4, the student's ability exceeds the difficulty of the question. At this point, the model predicts that the student would answer correctly, but the actual response does not match. This situation may be influenced by various factors, such as the student's random errors. This case illustrates that although the model generally predicts student responses well, occasional discrepancies such as in student S_1's e4 question arise due to random student errors, highlighting the model's limitations in accounting for such randomness.

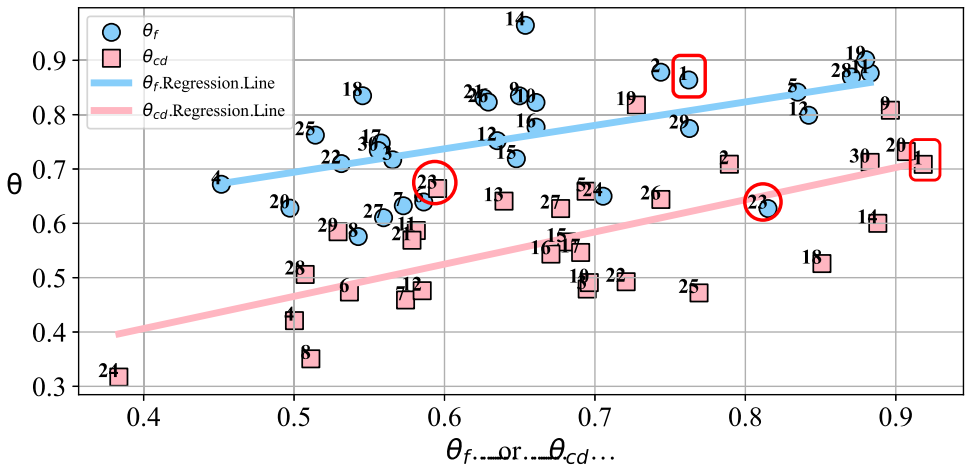


Fig. 7. Fusion of ability features for 30 randomly selected students from the science dataset, using average KC scores and Eq. (21). The higher the corresponding θ value for the two abilities of the same student, the greater the contribution.

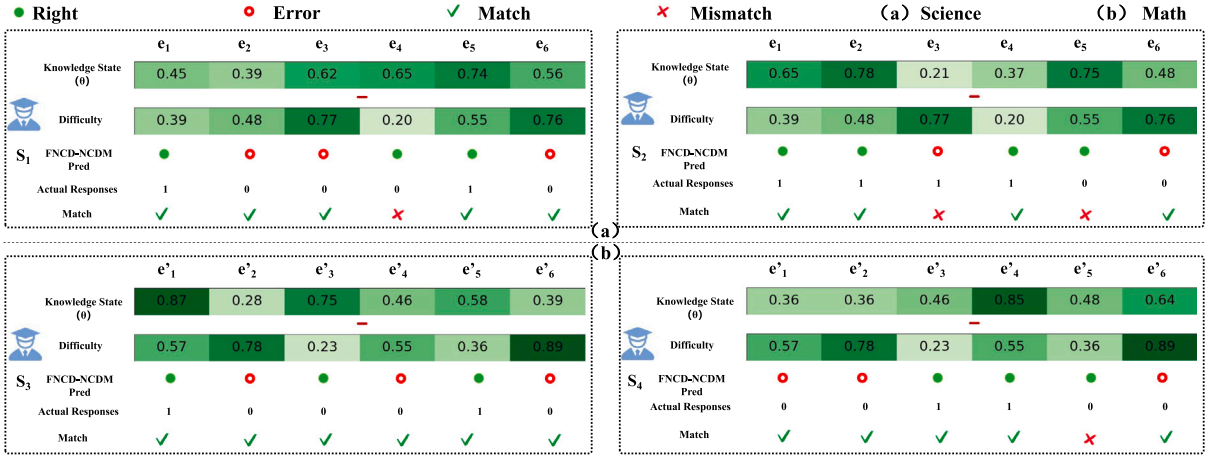


Fig. 8. FNCD-NCDM cognitive diagnosis case study..

For student S_2, in question e_3, the student's ability value is 0.21, while the question difficulty is 0.77. The model predicts that the student would answer incorrectly, but the actual response is correct. This phenomenon may be due to guessing during the student's response or indicate that there is still room for improvement in the model's ability to evaluate student capabilities. This example demonstrates that FNCD may underestimate student ability in certain cases, possibly due to guessing behavior or model evaluation limitations, suggesting further enhancement opportunities.

In contrast, the prediction for student S_3 in Figure (b) matches their actual response perfectly, indicating that with the introduction of fine-grained knowledge level constraints, the model can accurately assess students' knowledge level and make reasonable predictions in most cases. This case clearly shows that by integrating fine-grained knowledge constraints, FNCD achieves improved accuracy in cognitive diagnosis compared to traditional models, better reflecting students' knowledge mastery. Overall, these cases validate the FNCD model's strengths in correcting underestimations seen in traditional cognitive diagnosis models and enhancing interpretability and prediction accuracy.

6. Conclusions and limitations

This paper presents a novel fine-grained knowledge level constraint-aware diagnostic framework (FNCD), designed for the quantitative analysis and interpretation of students' cognitive state. We have developed a solution comprising three stages to systematically characterize students' knowledge mastery state and their group and individual characteristics.

In the KPA stage, we construct the Q-matrix and student response data and use a binary scoring-based average scoring method to estimate students' prior KC mastery level. Subsequently, in the BSS module, we introduce a random grouping strategy to divide students into multiple subgroups and use an attention mechanism to dynamically analyze intra-group similarities, constructing association edges among students. Finally, in the MSRL module, we design a multi-scale top-k heterogeneous graph neural network (AEGN) that enables dynamic feature interactions with each student's top-k similar peers, ensuring efficient interaction and reducing algorithm complexity while maintaining global feature integrity.

Based on this framework, we propose eight specific models: FNCD-IRT, FNCD-MIRT, FNCD-DINA, FNCD-NCDM, FNCD-AKT, FNCD-RCD, FNCD-TKT_CD, and FNCD-MHCD. A comparative analysis is conducted with the hierarchical constraint-based HCD model. Through extensive experiments on real educational datasets, we validate the effectiveness and interpretability of the FNCD framework. Furthermore, we conduct an in-depth analysis of the fine-grained knowledge level constraints of students within the FNCD framework, exploring their impact on group and individual knowledge state assessments, with the hope that this study will provide new insights for future CD research.

Despite these significant achievements, we acknowledge that the prior statistical methods in fine-grained knowledge level modeling remain coarse and cannot fully capture the complexity of students'

cognitive state. The learning process of students is inherently a multidimensional dynamic process, and modeling based solely on student ability features and question attributes cannot comprehensively reveal the deep connections between knowledge mastery and cognitive level. These issues present important directions for future research, particularly in the areas of more detailed prior knowledge modeling and dynamic CD.

CRediT authorship contribution statement

Xinjie Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Formal analysis, Data curation. **Qi Liu:** Resources, Funding acquisition. **Kai Zhang:** Validation, Funding acquisition. **Weiyin Gong:** Visualization. **Shuanghong Shen:** Resources, Data curation. **Fei Wang:** Supervision, Resources. **Yan Zhuang:** Data curation, Conceptualization. **Meikai Bao:** Resources. **Shijin Wang:** Funding acquisition, Formal analysis, Data curation. **Yuling Ma:** Funding acquisition, Data curation. **Enhong Chen:** Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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